

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 6

1. Estimation

- (a) Parametric estimation: random sample $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} f(x; \theta)$. f known, θ unknown. Target is to construct a point estimator $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ to estimate θ .
- (b) Bias and unbiased estimator: $\hat{\theta}$ is an unbiased estimator of θ if $\mathbb{E}_{X_1, \dots, X_n}[\hat{\theta}] = \theta$. If $\mathbb{E}[\hat{\theta}] \neq \theta$, $\mathbb{E}[\hat{\theta}] - \theta$ is the bias of the estimator.
- (c) Evaluation of point estimator
 - Bias-and-variance decomposition: mean squared error $\mathbb{E}[|\hat{\theta} - \theta|^2] = |\mathbb{E}[\hat{\theta}] - \theta|^2 + \mathbb{V}(\hat{\theta}) = \text{bias}^2 + \text{variance}$.
 - How can we interpret the mean squared error: average $|\hat{\theta} - \theta|^2$ over T simulations (for large T).
 - Minimum Variance Unbiased Estimator: Choose the one with minimum variance.
- (d) Some examples
 - Sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, sample variance $\hat{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$: both unbiased estimator.
 - (MLE estimator) Gaussian mean $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$, Gaussian variance $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2$.
 - (MLE estimator) Exponential parameter $\hat{\lambda} = n / \sum_{i=1}^n X_i$.

2. Maximum Likelihood Estimation

- Intuition: MLE sets the parameter so that the likelihood of seeing the data we observe is the largest.
- Procedure:
 - (a) Write down the likelihood function using the definition of the random sample and $f(x; \theta)$.

$$L(\theta; X_1, \dots, X_n) = p(X_1, \dots, X_n; \theta) = \prod_{i=1}^n f(X_i; \theta).$$

$L(\theta; X_1, \dots, X_n)$ is the function of θ .

- (b) It might be easier to work with the log-likelihood function (also a function w.r.t. θ)

$$\ell(\theta; X_1, \dots, X_n) = \log p(X_1, \dots, X_n; \theta) = \sum_{i=1}^n \log f(X_i; \theta)$$

- (c) Find $\hat{\theta}$ that maximizes $L(\theta; X_1, \dots, X_n)$ or $\ell(\theta; X_1, \dots, X_n)$ (the two are equivalent).
 - i. A standard routine: (1) calculate the first derivative of $\ell(\theta)$ with respect to θ , set it to be zero.
 - (2) show that $L(\theta)$ attains maximum value at some point.

- **Example 1 (MLE of Shifted Exponential)**

Consider n independent observations $x_1, \dots, x_n$ from the population with the shifted exponential pdf:

$$f(x; \lambda, \theta) = \begin{cases} \lambda e^{-\lambda(x-\theta)} & \text{if } x \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

The parameters θ and λ are unknown and we would like to obtain estimates. Note that θ, λ takes values in reals and $\lambda > 0$.

- (a) Obtain the maximum likelihood estimators of θ and λ .
 (Hint: First try to find the MLE of θ by considering what happens to the likelihood function when $\min(x_1, \dots, x_n) \geq \theta$ and $\min(x_1, \dots, x_n) < \theta$ separately. Then find the MLE of λ in terms of the MLE of θ , n , and $x_1, \dots, x_n$.)
- (b) If $n = 10$ with observed values 3.11, .64, 2.55, 2.20, 5.44, 3.42, 10.39, 8.93, 17.82, and 1.30, calculate the maximum likelihood estimates of θ and λ based on your results from the previous part.
- **Example 2** Suppose $X_1, \dots, X_n$ is a simple random sample from Uniform $[0, 2\theta]$ with some unknown parameter θ .
 - (a) Note that $\mathbb{E}[X_1] = \theta$. Then it coincides with estimating the mean parameter of the distribution. Consider the mean estimator $\bar{\theta} = \frac{1}{n} \sum_{i=1}^n X_i$. Is it an unbiased estimator? Find the bias and the variance for that estimator.
 - (b) Find the MLE for θ . Is it an unbiased estimator? Calculate the bias and variance for the MLE estimator $\hat{\theta}$.
 - (c) Which estimator is better (in terms of mean squared error)?